

Intelligent System for Real-time Health Monitoring of Power MOSFETs

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ABSTRACT

As Power MOSFETs play a significant role in the industrial manufacturing process, it is of immense value to study the failure modes of these MOSFETs. A typical MOSFET failure condition is caused by a short circuit between the source and drain. In addition, excessive current and excessive power dissipation are causes of MOSFET failure. The focus of this research was to create an intelligent data-driven system that uses Prognosis, Neural Networks, and Machine Learning Models to forecast the Remaining Useful Life (RUL) of power MOSFETs. After executing the train and test models on LSTM using a learning rate of 0.001, 50 epochs, and the Rul activation function, it was discovered that the MAE and model loss decreased substantially. A modest hardware setup was developed later for real-time testing of trained and tested LSTM models. After implementing data driven techniques, real-time testing was done under accelerated aging at 70°C. After 3 minutes of intensive accelerated aging, the MOSFET had 34.706 minutes of RUL. Future work will focus on developing a benchmarking circuit capable of operating under a wide range of temperature-controlled conditions. This will allow us to compare the activation energy derived from current models with that of the proposed model.

KEYWORDS: RUL, Prognosis, Predictive Health Maintenance, Remaining Useful Life, End-life Prediction

1. INTRODUCTION

Power MOSFETs are the key enabler of electrical gadgets. We all take for granted many modern electronics, from light-emitting diode lamps to high-definition televisions and digital cameras. They work by controlling electricity, much as a water valve controls water flow. Apart from these applications, power MOSFETs are widely used in the automotive industry. Electric bikes are a great example. This is borne out by the fact that no electronic device works without a MOSFET somewhere inside it. Power MOSFETs is the only MOSFET that makes high voltage very accessible for microcontrollers. A normal MOSFET cannot switch high voltages due to the dielectric breakdown of the gate insulator damaging the gate. The use of new manufacturing methods has led to a decrease in the thickness of gate oxide, which has led to an increase in dynamic characteristics and reduced power loss. The MOSFET may get overheated or get radiation, both of which will cause the oxide-trapped charge and interface traps to be considerably denser [1]. This causes instability in the system. Therefore, researchers have paid much attention to the failure problems of power MOSFETs in the past years. It's also important to note that bond wire-related failures like cracks and delamination in the die attach solder are considered one of the causes of increased on-state resistance. A typical failure condition in a Power MOSFET is a short circuit between source and drain.

Due to the increased demand for safety and maintenance requirements in industrial applications, prognosis has gained great importance in the last few decades. This makes the industry preplan its maintenance by adopting agility. In this project, an LSTM model was employed as a machine learning tool to be able to measure and predict the RUL of the power MOSFETs. The key results stated that the resistance of the junction of the MOSFETs is the main key element for its prognosis. The train, test, and real-time data sets each validated the response from the NASA datasets.

A. MONITORING METHODS AND ALGORITHMS

Various health monitoring methods have been employed to predict the degradation of power MOSFETs in previous studies. Health monitoring methods based on the threshold voltage [2], condition monitoring method for threshold voltage drift and packaging degradation [3] are some of the methods used for health monitoring of power MOSFETs. These approaches offer valuable insights into the operational reliability of power MOSFETs under stress. Several algorithms have been developed to enhance the accuracy of power MOSFET health monitoring, including:

- Particle-Based Filter Method [4]
- Platform Voltage Method (Miller Effect) [1]



Figure 1. Stepwise processes in the calculation of RUL

- Rain flow Counting Algorithm [5]
- Kalman Filter Method [5]
- Feed Forward Neural Network Mechanism [6]

These methods are particularly discussed in the 'State-of-the-art' section of this paper.

2. STATE OF THE ART

Power MOSFETs RUL can be predicted using prognostics, according to Celaya et al. [7] in 2011. Thermal cycling and Gaussian process regression were utilized for prediction. Temperature-dependent changes in ON-state resistance was employed as a warning sign that the device was about to fail. In order to make the proposed procedure work, several assumptions were made. During the accelerated aging process, the loads and conditions that would be applied to the device in the future were assumed to be constant. According to the findings, as more data becomes available, the predictions get more precise (as evidenced by the proximity of the predicted dashed lines to the crossing of the failure threshold by the ON resistance) and the prediction range becomes more precise.

Power MOSFETs that are subjected to electrical overstress at the gate area are subject to rapid aging, as proposed by Saha et al. [8] in 2011. The methods for accelerating the aging process, modeling the device degradation process, and a prognostics algorithm for predicting the future health of the device were all covered in the paper's presentation of the findings. In a particle filtering system, the degradation curve model was applied to predict RUL estimations. Although the prognostics routine was solid, it could not deal with aspects such as irregular rest intervals that occur with age as part of the natural process.

In 2014, Zheng et al. [9] studied to estimate the RUL of power MOSFETs, a technique merging RVM, and a deterioration model was suggested in this research. To begin, RVM used available power MOSFET deterioration data as training data to generate RVs. Secondly, using representative vectors and nonlinear least squares regression, the parameters of the deterioration model were determined. Then, by expanding the model of degradation to a failure threshold, the RUL of the power MOSFETs was calculated. Finally, throughout the prediction phase, the degradation model was updated according to the recommended approach. As a result, the model was able

to respond to the mutation promptly, and the prediction became more accurate.

Dusmez et al. [10], 2015, studied the Remaining Useful Life (RUL) of power MOSFETs which are degraded by the thermal cycling process. A data-driven approach is used in which data is collected through exhaustive tests. An exponential curve fitting algorithm was used to find empirical coefficients α, β online. The uncertainties such as noise errors were eliminated by using Kalman Filter (KF) method. Only on-state resistance parameters were observed as junction temperature measurements were not required in this model. It was concluded that the initial estimation errors, for exponential coefficients α, β , were reduced to 4.2% and 3.5% from 34.6% and 28.7%, by using more data points.

In 2017, Dusmez et al. [11] analyzed the degraded power MOSFETs failure due to thermal cycling. A data-driven approach RUL estimation algorithm was used, and the errors were removed by Random Sample Consensus (RANSAC) algorithm. In comparison to the case where the least-squares method is applied to all data points without removing errors, the proposed methodology yielded superior results to the actual RUL. The use of RANSAC ensures that the estimation is resistant to large levels of noise

In 2017, Chen et al. [1] devised an in-situ prognosis method for the study of power MOSFETs based on the miller effect. A new degradation precursor the miller platform voltage was identified. Algorithms like particle filters were used to predict the RUL of MOSFET using miller voltage data. It was concluded that degradation of power MOSFET is increased by increasing the gate bias voltage. Also, the degradation trend of miller platform voltage matched with a threshold voltage.

In 2018, Pugalenth et al. [4] investigated the prognosis of power MOSFET microcontroller devices. They claimed that power MOSFETs degrade owing to thermal reasons, and they used ON-resistance variation as a failure indicator for prognosis. They adopted the particle filter algorithm for their purpose. They discovered an improvement in accuracy above the usual particle filter method towards the end of their analysis. The effect of varied roughening on RUL prediction accuracy is discussed and the results are obtained by using the parameters which were roughening to get knowledge such as comparative accuracy and computational period.

In 2021, Witczak et al. [12] predicted the remaining useful life of power MOSFETs using analytical and datadriven methods. In the proposed approaches they obtained the degradation data from the historical data. As a datadriven technique, they employed the Takagi-Sugeno multiple models-based frameworks, while as an analytical approach, they used an exponential curve fitting online approach. Unlike the previous prognosis, they designed data on the basis of historical data to get the diagnostics decisions. It was concluded that the currently operating MOSFETs, went through the run-to-failure process. Finally, the suggested method is validated using real data obtained from NASA's prognostic data collection.

This paper presents a solution to this problem by proposing a machine learning approach for predicting the RUL of Power MOSFETs. The proposed model is based on LSTM and is an iterative process that uses a regression LSTM to estimate the error between the actual data and predicted values for each sample in the dataset. The final prediction is obtained by taking the average of all these estimates. The novelty of this work shows intelligent data-driven system will calculate the RUL of individual power MOSFETs, with minimum errors of MSE; RMSE; and MAE. Real-time model testing will be done on MOSFET to ensure its useability in industries.

3. METHODOLOGY

In this section, a comprehensive description of the proposed materials and methods is presented, along with a description of the complete experimentation that will be used for this work. Listed below are the different subdivisions of this section with detailed information about work and outcomes from this work.

A. DATA SETS

Model degradation datasets were acquired from the NASA Ames Prognostics Repository. The data consisted of 42 MOSFETs data which were run-to-failure under thermal stress. The dataset includes detailed measurements of the MOSFETs' electrical and thermal characteristics over time, providing insights into their degradation patterns. Various environmental factors, such as temperature swings, are applied to accelerate aging and observe the impact on performance. The wide temperature ranges from (-55 to +175°C). The format data was available in .mat format, in which transient response, steady-state response, and PWM data were formulated into multiple ensemble tables.

B. MODERN TOOLS USED

- MATLAB: Used for pre-processing and data analysis

- Jupyter-Labs: Environment for Python and Machine Learning
- Python: Keras, Pandas, Seaborn, NumPy, and TensorFlow libraries used
- Data Streamer: Used to extract real-time data

1. MATLAB

At primary the current and voltage values were available in the form of ensemble tables. Using the average values of the ensemble table data, such as MOSFET drain current and drain-to-source voltage, the data at one timestamp couldn't give noise in the values. Since we know that the RUL of MOSFETs is mostly dependent upon resistance, we used to drain current and drain-to-source voltage to determine the resistance. After the separation of all the variables in a CSV file, filters like median filter were applied to the resistance data to overcome the noise. Figure 2 shows the difference between the raw resistance and filtered resistance of the 26th MOSFET.

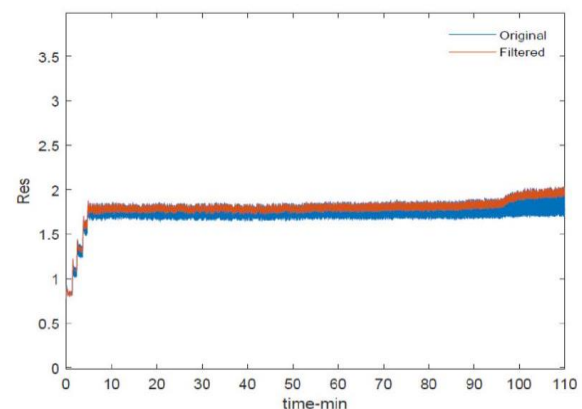


Figure 2. Filtered R using a median filter of the 26th MOSFET

2) Python

The results of the MATLAB algorithms were then used to carry out additional processing. We took the average values of 25 and 75 preceding data points to remove the spiky data. The reduction of background noise and data cleansing allowed the machine learning model to run more quickly and with fewer errors than previously. Data normalization was also performed on the data which rescaled the values to 0 and 1. Model training becomes less dependent on the scale of features when data normalization is utilized. Figure 3 shows the resistance values of the 17th MOSFET which was then filtered using the mean average filter in the python environment as shown in Figure 4.

C. CONDITION MONITORING

It was necessary to perform conditional monitoring to identify the factors that might affect the prognosis. The drain current reduces as the drain-to-source voltage rises over time, according to the results of the experiment. Because voltages and currents affect resistance, we can simply use the resistance values to get accurate results from the model. This means that the model can be implemented in a fraction of the computing time it would require if it had more

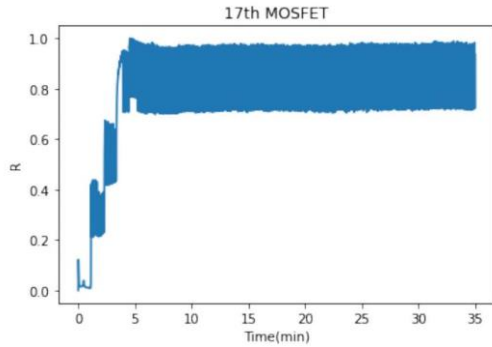


Figure 3: Resistance of the 17th MOSFET (normalized)

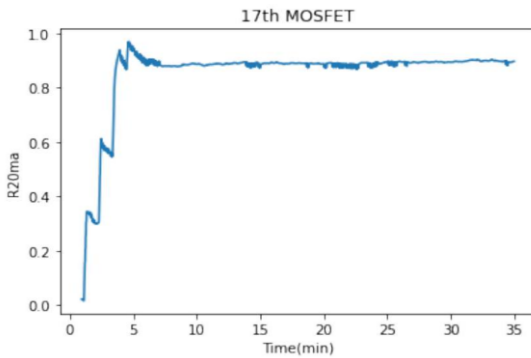


Figure 4: Filtered Resistance of the 17th MOSFET (normalized)

variables. Figure 5 depicts the relationship between the 26th MOSFET's R, VSD, and DC. This graph shows that all three variables are linked together. In this equation, R stands for resistance, VSD for drain-to-source voltage, and DC for MOSFET drain current. The graphs obtained from the conditional variable's identification are shown in Figure 6. From Figure 6, it can be seen that the resistance, drain current and the drain-to-source voltage along with the independent variable "time" will help calculate the RUL of the MOSFET.

Figure 7 shows the relationship between the filtered values of resistance of the three MOSFETs numbered 26, 17, and 21. It can be seen from the graph that the resistance of all the three MOSFETs is different from each other even if the operating conditions are the same for all of them.

D. MACHINE LEARNING MODELS

For the prediction of power MOSFETs, LSTM was used as a machine learning model. Due to its accuracy and ability to operate well with time series data, it was chosen. The long-term reliance problem is deliberately avoided with LSTMs.

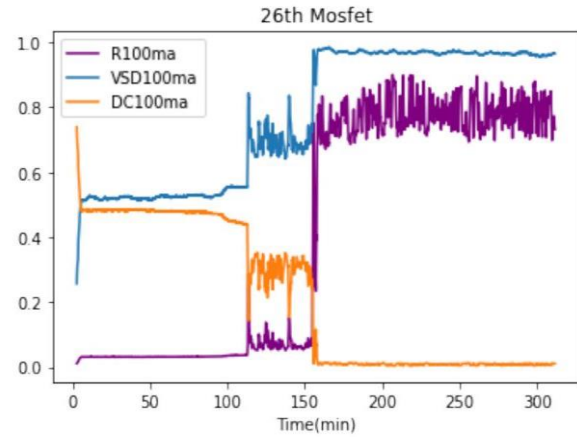


Figure 5: Resistance, VSD, and DC of the 26th MOSFET

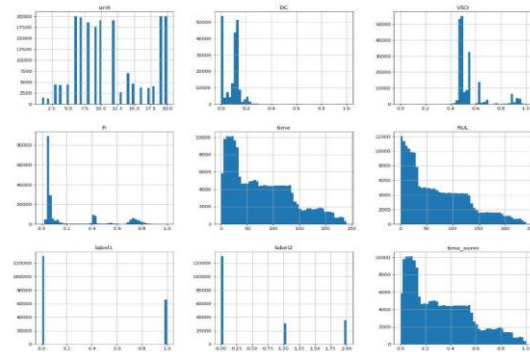


Figure 6: Dependent and independent variables for conditional indicators

For them, long-term memory is a natural situation, and they don't have to work at it. The RUL activation function was used with 30 epoch values in the LSTM's sequential model. The MSE (mean squared error) dropped with each occurrence of the following epoch value when the model was run under 30 epochs. 0.001 learning rate was adopted for training on LSTM.

E. RUL PREDICTION

RUL, to train the model, was calculated by using Eq 1 .

$$MTD - CTPI \quad (1)$$

Where;

MTD = Max time of degradation

CTPI = Current Time at the present instance

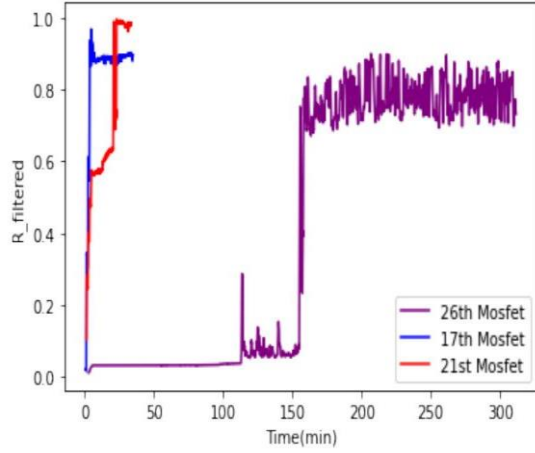


Figure 7: Filtered resistance values of 26, 17, and 21 MOSFET

Afterward, a health assessment of the power MOSFET was done by training the machine learning model, calculating the errors, and testing it with the remaining data.

F. REAL-TIME MONITORING SETUP

For the real-time testing of the trained and tested LSTM models, a small hardware setup was designed. The components include Voltage Measuring Sensor, Current Measuring Sensor, MOSFET 80nf70, Motor, Arduino Mega, Breadboard, Jumper wires, 5V Power Source, and Excel Data Streamer.

Excel's Data Streamer is a two-way data transmission that transfers data from Excel back to a microcontroller, as well as streaming live data into Excel from a microcontroller. Students are introduced to the fields of data science and the internet of things through the use of Excel, sensors, and microcontrollers that are included in the Data Streamer kit (IoT) [13]. Arduino was sending out data in the form of VSD and DC. Using Excel Data Streamer, these were saved in the format of tables as a comma-separated values (CSV) file. The raw data was pre-processed similarly as the training historical data and was normalized to ensure efficient model implementation. This data was then utilized to compute the resistance of the MOSFET, which, according to ohm's law, is the only condition indicator for the RUL. This value was found using Eq 2

$$R = \frac{V}{I} \quad (2)$$

4. RESULTS AND DISCUSSION

A. HEALTH ASSESSMENT

Table IV-A presents the LSTM model's MSE, RMSE, and MAE losses. Because of this, it can be concluded that the model has been properly trained and that its accuracy has increased over time. Using the MSE, researchers determine how closely a fitted line matches the data. The better the model fits the data, the lower the Mean Squared Error will be. Error rate divided by square root of MSE is RMSE. This statistic is the simplest to understand because it has the same units as the quantity depicted on the Y-axis or vertical axis.

Table 1. Error and Loss values of the trained LSTM Model

Error/Loss Type	Values
Mean Squared Error (MSE)	9.615992022829877
Root Mean Squared Error (RMSE)	3.1009663046911484
Mean Absolute Error (MAE)	2.340416395825946

$$MSE = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2 \quad (3)$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^N (x_i - \hat{x}_i)^2}{N}} \quad (4)$$

The health assessment of the MOSFET, after its training, was accomplished by testing the model and pointing out the maximum errors in them. Data of the 20 MOSFETs were compared with each other. The testing model was run to fit the trained model of all the 20 MOSFETs.

After running the train and test models on LSTM, with configurations discussed in the methodology section, it was found that the MAE and model loss dropped significantly. This loss and MAE can be seen in Figure 8 and Figure 9. From the two figures, it is clear that running the learning algorithm within 50 epochs provides the best results for MAE and loss function. Changing the epoch values will result in overfitting or underfitting respectively. The model MAE in Figure 9 would stop at 4.6, and it would not decrease further. The train and test sets of the fitted data is almost same as shown in the figure. However, the test data MAE shows a further decrease in trend as compared to the training data. To put it simply, the test set was easier for the model to do well on than the training set, which resulted in a lower Test MAE!

Training and testing RUL data were plotted on a graph after losses had been calculated. It was obtained by fitting

the LSTM model with previously defined y-prediction values. It can be seen from the Figure 10 that the anticipated and real RUL for the training RUL data is nearly identical. An accurate test on historical and real-time data confirms that the model has been properly trained and can be moved along to test on the historical and real-time data. Similarly, in Figure 11, for the testing RUL predicted vs actual RUL data, the graphs also show the identical trend.

B. REAL-TIME PREDICTION

As discussed in the earlier sections, a real-time circuit was developed to validate the trained and the tested LSTM models in order to ensure the model inclusiveness. The same pre-processing techniques were applied as used for NASA Mosfet Degradation dataset.

The real-time tests were run for 3 minutes under highly accelerated aging process with the operating temperature of almost 70°C. The derived results from the machine

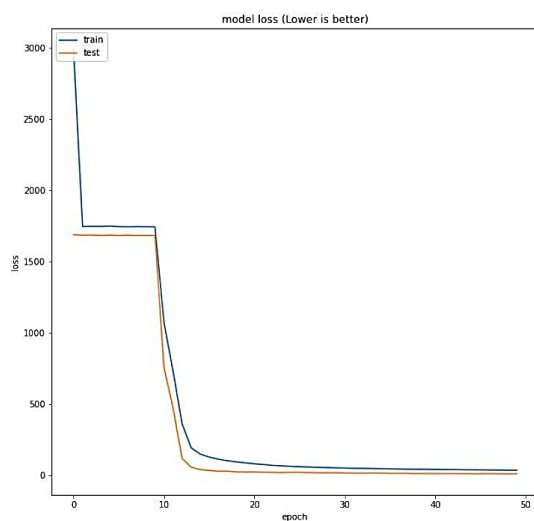


Figure 8. Model loss of the train and test data

RUL with minimal error, achieving MSE of 9.615, RMSE of 3.101, and MAE of 2.340. The LSTM-based model demonstrated significant improvements over traditional methods, particularly in handling time-series data and providing accurate long-term predictions.

Our approach offers practical implications for real-time health monitoring in industrial settings, enabling predictive maintenance and reducing the risk of unexpected power MOSFET failures. The real-time validation of the model under accelerated aging conditions confirmed its application for real-world scenarios.

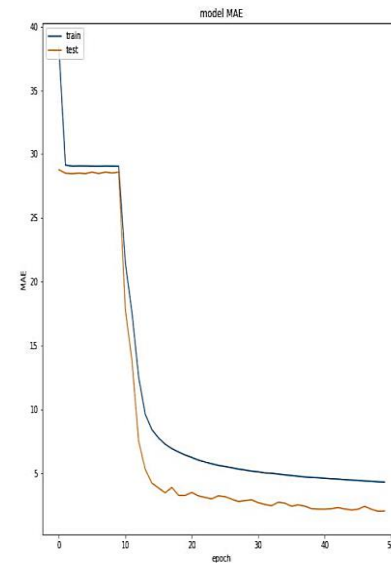


Figure 9. Model MAE of the train and test data learning model showed that after 3 minutes of intense accelerated aging, the MOSFET remaining useful life was 34.706 minutes. The derived results for the real-time RUL is shown in figure 10.

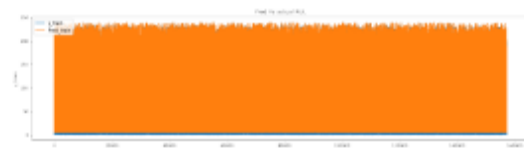


Figure 10. Original and Predicted values of the training dataset

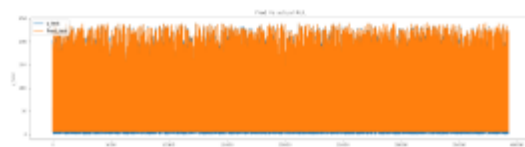


Figure 11. Original and Predicted values of the training dataset

While the results are promising, future work should focus on testing the model under a wider range of operating conditions to improve its robustness. Additionally, the development of a bench-marking circuit for comprehensive testing across varying temperatures will further enhance the accuracy and applicability of the proposed method.

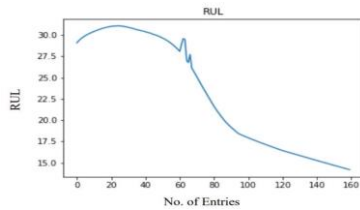


Figure 12. Real-time RUL of a power MOSFET

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